

SKILL WEEK-1

Session 1

Task1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Make file.

Outputs:

```
make: Nothing to be done for 'all'.
sathwika@123:~/ossp_2520090210/skill/Week1/Task1$ ./main
Hello, OSSP Skill Week 1!
GCC compilation is working successfully.

sathwika@123:~/ossp_2520090210/skill/Week1/Task1$ nano main.c
sathwika@123:~/ossp_2520090210/skill/Week1/Task1$ gcc main.c -o main
sathwika@123:~/ossp_2520090210/skill/Week1/Task1$ ./main
Hello, OSSP Skill Week 1!
GCC compilation is working successfully.
```

Verifying linux environment by uname -a:

```
sathwika@123:~$ uname -a
Linux sathwika 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

Checking GCC C compiler:

```
sathwika@123:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Git version:

```
sathwika@123:~$ git --version
git version 2.53.0
```

Observation: I learned how to work with the Linux environment and verify system tools such as GCC and Git. I learned how to compile and execute a C program using GCC and understood the purpose of a Makefile for automating the compilation process. I also learned how Linux development projects can be organized and built using standard compiler and build commands.

Task2: To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, practice system call tracing.

Output:

```
sathwika@123:~$ nano skill_week1_task2.c
sathwika@123:~$ gcc skill_week1_task2.c -o skill_week1_task2
sathwika@123:~$ ./skill_week1_task2
Parent Process Started
Parent PID: 507

Parent Process
Parent PID: 507
Child PID: 508
Parent waiting for child...
```

```

Child Process
Child PID: 508
Parent PID: 507
Executing ls using exec()
total 248
-rw-r--r-- 1 sathwika sathwika 109 Aug 7 05:17 File1.txt
-rw-r--r-- 1 sathwika sathwika 109 Aug 7 05:19 File2.txt
-rw-r--r-- 1 sathwika sathwika 0 Aug 5 03:41 LICENSE
-rw-r--r-- 1 sathwika sathwika 0 Aug 5 03:41 Makefile
drwxr-xr-x 4 sathwika sathwika 4096 Aug 11 06:30 OSSP
-rw-r--r-- 1 sathwika sathwika 0 Aug 5 03:41 README.md
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 assets
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 bin
-rwxr-xr-x 1 sathwika sathwika 16528 Aug 13 06:59 command_executor
-rw-r--r-- 1 sathwika sathwika 1387 Aug 11 06:59 command_executor.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 docs
-rwxr-xr-x 1 sathwika sathwika 16216 Aug 7 05:19 file_copy
-rw-r--r-- 1 sathwika sathwika 1113 Aug 7 05:17 file_copy.c
-rwxr-xr-x 1 sathwika sathwika 16432 Aug 13 08:28 fork_exec
-rw-r--r-- 1 sathwika sathwika 947 Aug 13 08:28 fork_exec.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 include
-rwxr-xr-x 1 sathwika sathwika 16192 Aug 7 05:39 keyboard_input
-rw-r--r-- 1 sathwika sathwika 1803 Aug 7 05:39 keyboard_input.c
-rwxr-xr-x 1 sathwika sathwika 16352 Aug 7 05:09 main_loop
-rw-r--r-- 1 sathwika sathwika 800 Aug 7 05:09 main_loop.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 obj
drwxr-xr-x 3 sathwika sathwika 4096 Aug 11 06:41 ossp_2520090210
-rwxr-xr-x 1 sathwika sathwika 16824 Aug 11 06:41 parser_double_quotes
-rw-r--r-- 1 sathwika sathwika 3934 Aug 11 06:41 parser_double_quotes.c
-rwxr-xr-x 1 sathwika sathwika 16560 Aug 11 06:34 parser_quotes
-rw-r--r-- 1 sathwika sathwika 2249 Aug 11 06:34 parser_quotes.c
-rwxr-xr-x 1 sathwika sathwika 15952 Aug 1 04:49 process
-rw-r--r-- 1 sathwika sathwika 80 Aug 1 04:48 process.c
-rw----- 1 sathwika sathwika 80 Aug 1 04:46 process.c.save
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 scripts
-rwxr-xr-x 1 sathwika sathwika 16304 Aug 13 10:14 skill_week1_task2

-rwxr-xr-x 1 sathwika sathwika 16304 Aug 13 10:14 skill_week1_task2
-rw-r--r-- 1 sathwika sathwika 879 Aug 13 10:13 skill_week1_task2.c
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 src
drwxr-xr-x 2 sathwika sathwika 4096 Aug 5 03:41 tests
Child process completed.

```

Observation: I learned how Linux creates and manages processes using fork(). I understood that exec() can replace the child process with another program, while wait() allows the parent process to wait until the child process finishes.

Session 2

Task1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

Output:

```

sathwika@123:~$ cd ~
sathwika@123:~$ nano session2_task1.c
sathwika@123:~$ gcc session2_task1.c -o session2_task1
sathwika@123:~$ ./session2_task1

```

```
OSSP Interactive Shell
myshell> hello
You entered: hello
myshell> test
You entered: test
myshell> exit
Exiting shell...
```

Observation: I learned how to create an interactive main loop in C that repeatedly displays a prompt and reads user input. I also learned how to handle exit conditions, manage input strings, and control program flow using loops and conditional statements.

Task2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

Output:

```
sathwika@123:~$ nano session2_task2.c
sathwika@123:~$ gcc session2_task2.c -o session2_task2
sathwika@123:~$ ./session2_task2
Type a command and press Enter:
input> hello world

Command entered: hello world
Characters stored: 11
```

Observation: I learned how to capture keyboard input in C using `getchar()`. I understood how to manage an input buffer to store multiple characters, process the Enter key to finish input, and handle the Backspace key by removing the last character from the buffer. I also learned how user input can be processed and displayed as a complete command.